PU WANG

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EDUCATION

College of Computer Science and Technology, Zhejiang University **Hangzhou, China**
B.E. in Artificial Intelligence (Turing Class) *Sep 2023 – Jun 2027 (expected)*
Honors degree from Chu Kochen Honors College
Related courses: *Introduction to Theoretical Computer Science, Introduction to Natural Language Processing, Machine Learning, Mathematical Analysis I/II*

RESEARCH INTERESTS

My research interests lie in the theory and algorithms of **sequential decision making**, with a growing focus on **large language models** and **language agents**. I am interested in how agents acquire, through interaction, the knowledge they need to act well, and how that knowledge can be learned efficiently and reused across tasks and environments. More broadly, while algorithms and theory remain at the core of my work, I aim to bring principled formulations from bandits and reinforcement learning to problems related to decision making in language models and language agents.

PUBLICATIONS

RoleWorld: Knowledge Transfer in World Modeling with Functional Role Abstractions.
Boyuan Xiao, ***Pu Wang****, Ji Feng, Yao-Xiang Ding*
Preprint: arXiv:2610.13172

Tree-Guided Identify-Then-Exploit: A Unified Framework of Best Arm Identification and Regret Minimization for Dueling Bandits.
Pu Wang*, Yao-Xiang Ding*
NeurIPS 2026*: The Fortieth Annual Conference on Neural Information Processing Systems*

RESEARCH EXPERIENCE

Research Intern at State Key Lab of CAD&CG Mar 2025 – Present
Advisor: Prof. Yao-Xiang Ding Zhejiang University

- **TG-ITE: A Unified Framework for Dueling Bandits** *Mar 2025 – May 2026*
  Developed the first unified framework for N-armed dueling bandits that jointly handles best arm identification (BAI), weak regret, and strong regret under only the Condorcet-winner assumption. It achieves $O(N)$ sample complexity for BAI and $O(N)$ weak regret, and **removes the** $O(\log N)$ **suboptimality gap** of the existing approach when optimizing both jointly.
- **Imitation Learning from Partially Observable Experts** *Jun 2025 – Dec 2025*
  Studied imitation from multiple experts that each observe only a masked part of the state. Proposed **POIL**, a GAIL-based method with active querying, with preliminary MuJoCo experiments.
- **RoleWorld: Knowledge Transfer in World Modeling** *Jun 2026 – Sep 2026*
  Studied how world models of LLM agents can transfer to new environments. RoleWorld represents how objects behave with **functional role abstractions**: reusable role prototypes with environment-specific parameter values. In a new environment, the actor LLM refines role assignments and parameter estimates online from action feedback, without LLM parameter updates.

PROJECTS

Needle: A Compiler from C to RISC-V Mar 2025 – Jun 2025
Course Project for Principles of Compilers Zhejiang University

- **Frontend & IR:** Implemented an end-to-end compiler for a C subset using *Flex* and *Bison*, featuring AST construction, semantic analysis, and Intermediate Representation (IR) generation.
- **Code Generation:** Developed a RISC-V code generator handling calling conventions and control flow, verifying correctness through simulation tools.

Linear Algebra: Left Undone
Open-Source Supplementary Lecture Notes Chu Kochen Honors College, Zhejiang University

- **Overview:** Open-source supplementary linear algebra lecture notes of Chu Kochen Honors College, Zhejiang University, with 700+ stars on GitHub.
- **Contribution:** Authored Chapter 24, *Linear Algebra and Calculus*.

ACHIEVEMENTS

Zhejiang University Scholarship – Third Prize *2026*
Zhejiang University Scholarship – Second Prize *2024, 2025*
Second Class Scholarship for Elite Students in Basic Sciences *2025*

SKILLS

Programming Languages	Python, C/C++
Machine Learning Tools	PyTorch, einx/einops, Gymnasium, Stable-Baselines3
Systems & Infra	Linux, Git, Shell Scripting, Docker, Slurm
Writing & Typesetting	LaTeX, Typst, TikZ

I maintain an open technical notebook (400,000+ words) on computer science, mathematics, and AI, as a long-term knowledge-building practice.